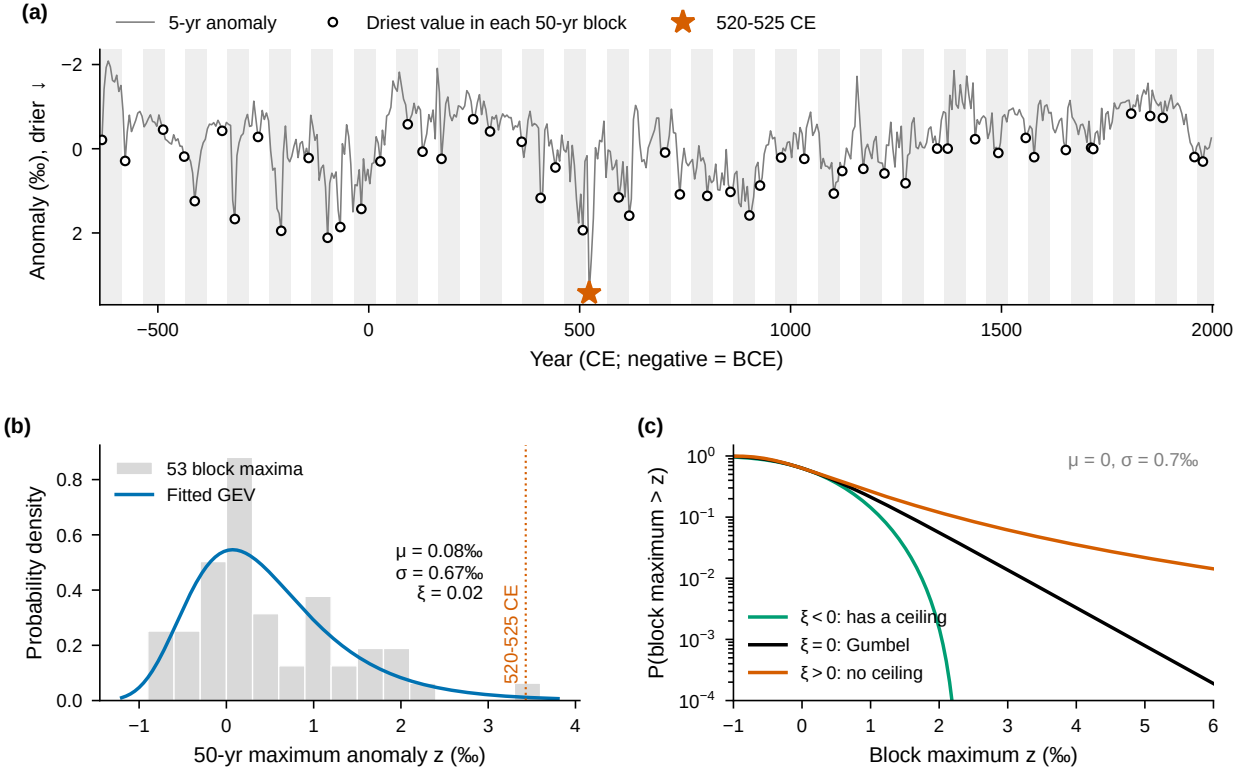
(a)
5-yr anomaly
Driest value in each 50-yr block
520-525 CE
Anomaly (‰), drier ↓
−2
0
2
−500
0
500
1000
1500
2000
Year (CE; negative = BCE)
(b)
53 block maxima
Fitted GEV
μ = 0.08‰
σ = 0.67‰
ξ = 0.02
520-525 CE
Probability density
0.0
0.2
0.4
0.6
0.8
−1
0
1
2
3
4
50-yr maximum anomaly z (‰)
(c)
μ = 0, σ = 0.7‰
P(block maximum > z)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
ξ < 0: has a ceiling
ξ = 0: Gumbel
ξ > 0: no ceiling
−1
0
1
2
3
4
5
6
Block maximum z (‰)